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Expectations and beyond: The nexus of AI instrumentality and brand credibility in voice assistant retention using extended expectation-confirmation model

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Abstract

AI-based voice assistants (AIVA) are capable of interpreting human speech and responding with useful information, aiding with tasks, and controlling other devices. The usage of these AIVAs has grown significantly worldwide. Despite this growth, studies on user behavior related to continued usage intention of AIVAs and effects on the long-term commercial sustainability of brands, remain low. What is less understood is the potential of AI instrumentality attributes and brand credibility components in provoking shifts in post-use behavior of AIVAs. This study proposes a model which expands on the Expectation-Confirmation Model for user continuance behavior of AIVAs, by integrating user's technology related traits, AI instrumentality attributes and brand credibility. To verify the research hypotheses, the study employed partial least square—structural equation modelling, based on 281 validated responses of a survey. This study highlights the significance of Optimism, Innovativeness, and Discomfort for post-adoption confirmation. Higher post-adoption confirmation is strongly associated with perceived intelligence, anthropomorphism, information quality, and system quality. Anthropomorphism and information quality are key factors for brand expertise, while anthropomorphism and system quality are significant for brand trustworthiness. The study confirms that brand expertise and trustworthiness lead to post-use satisfaction and use continuance intention. Understanding the antecedents of satisfaction and continuance intention extends the existing literature on AIVAs and provides valuable insights for academics and practitioners alike. Some implications for researchers and managers are discussed.

1 | INTRODUCTION

The global voice assistant market is expected to grow significantly at a compound annual growth rate of 24.32% from 2019 to 2025, with an estimated market size of nearly USD 7.3 billion by 2025 (Market Research Future, 2021). The use of voice assistants is expected to expand, particularly as they become increasingly valuable for branding, marketing, and customer loyalty (Moriuchi, 2019).

The term 'voice assistant' is commonly used interchangeably with other phrases such as digital assistant, chatbot, intelligent virtual assistant, interactive voice assistant and AI agent. AI-based voice assistant (AIVA), which belongs to the category of voice-enabled artificial

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intelligence, have become ubiquitous, with various companies introducing their own versions (Atieh, 2021).

Numerous studies have noted that interactions with AIVA can elicit emotional responses, which create a perception of personal relationships (Park & Namkung, 2021; Shank et al., 2019). Amazon's Alexa, Apple's Siri, Google Assistant, and other voice conversational AI, and screen-less interfaces have revolutionized the way users conduct basic personal tasks, interact with businesses (e.g., customer experiences), and utilize home automation (McLean & Osei-Frimpong, 2019).

Despite the significant growth in the use of AIVAs, studies on user behavior and their continued usage intention of AIVAs, and its impact on the long-term commercial sustainability of brands, remain scarce. Most studies on the use continuance intention of AIVAs, have focused mainly on user studies' traits, attitudes, or behavioral perspectives (e.g., Pal et al., 2021; Zhu et al., 2022).

The Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) are the most widely used theoretical models for studying the adoption of innovation including AIVAs. However, these models primarily focus on the factors that drive adoption and do not sufficiently address user satisfaction and its relationship to continued use. AIVAs rely heavily on user feedback to enhance their accuracy and efficacy. Users who interact with AIVAs establish expectations regarding how the assistant will behave, respond to inquiries, and provide assistance. Thus, the Expectation-Confirmation Model (ECM) was deemed suitable for investigating AIVAs since it allows researchers to evaluate how well the technology meets user expectations and how these expectations evolve over time.

Prior research on AIVAs has not thoroughly explored the specific product instrumentality attributes unique to AI, such as Anthropomorphism, Intelligence, Information Quality, and System Quality, and branding perspectives in understanding users' attitudes towards these assistants (Salles et al., 2020). Furthermore, the influence of brand credibility on user satisfaction, loyalty, and continuance commitment has been recognized in several studies (Dandis et al., 2022; Sweeney & Swait, 2008). It is necessary to further comprehend the potential of these antecedents in inducing changes in post-use behavior concerning AIVAs.

Previous studies have also indicated that user satisfaction plays a vital role in discouraging switching behaviors and increasing the retention rate (Mutum et al., 2014). Thus, this study aims to address these gaps in the literature by extending the ECM to investigate the impact of AI-instrumentality attributes, Brand expertise and brand trustworthiness on the continuance intention to use digital assistants. Our study aligns with the call by Lim et al. (2023) to explore consumer engagement with new-age technologies, such as AIVAs, not just from a technological perspective but also from behavioral, psychological, and other perspectives. By considering these factors, we can gain a better understanding of the drivers of user satisfaction and their potential role in encouraging continued use of AIVAs. The following section begins by laying out the theoretical background and hypotheses development. This is then followed by the research methodology, results and finally the conclusion.

2 | CONCEPTUAL, THEORETICAL FRAMEWORK, RESEARCH MODEL AND LITERATURE SUPPORT

Most technology acceptance and use studies have used TAM and UTAUT model in studying initial adoption intentions (Sohn & Kwon, 2020). However, initial adoption and continued use are built on different foundational empirical concepts. Continuance (user retention) is not just an extension of initial adoption (user acquisition), where significant factors from post-adoption usage are disregarded (Bhattacherjee, 2001). Moreover, in terms of post-adoption behaviors of AIVAs, most studies focus largely on purely personal traits, attitude, or behavioral perspective of a user (e.g., Lee & Sheehan, 2021; Zhu et al., 2022). Not much consideration has been given on product instrumentality (specific to AI) and branding perspectives to understand users' disposition towards AIVAs (Jain et al., 2022).

Some studies have answered the call by Lim et al. (2023) to study new-age technologies such as AIVAs from the behavioral, psychological, and other perspectives, for example, Lee and Park (2022) used parasocial theory to examine how interactions with AIVAs have an impact on consumers' perceptions of communication quality, satisfaction, and ultimately on continuance usage intentions. The current study builds on their findings by looking at the use continuance intention of AIVAs from a slightly different theoretical lens, namely, the ECM, which builds on the self-perception theory put forward by Bem (1972). It posits that after the initial acceptance of technology, a user's satisfaction, and perception of its usefulness (a cognitive assessment of its instrumentality) may change. Thus, users may either abandon or repeat the behavior. ECM consists of four constructs: post-use confirmation of expectations, perceived usefulness, post-use satisfaction, and user continuance intention (Bhattacherjee, 2001). However, the original ECM was not considered sufficient to explain the use continuance intention. This study extends the model to include AI and instrumentality attributes (Anthropomorphism, Intelligence, Information Quality, and System Quality), as well as brand credibility attributes (Brand Trustworthiness and Brand Expertise), which are likely to be contingent in the context of AIVAs. These are discussed in more detail in the hypothesis's development section.

2.1 | Hypotheses development

According to the ECM, users' perceived usefulness and satisfaction (post-use) are directly influenced by their confirmation of expectations (pre-use) about the benefits of using technology (Mamun et al., 2020). Confirmation of pre-acceptance expectation is a crucial differentiator for ECM, as it involves the considerations of satisfaction and confirmation, which are critical factors in describing post-use behaviors (Oghuma et al., 2016). The confirmation of expectations affects continuance intention to use directly and indirectly through confirmation of expectations (Meidani et al., 2022). Mamun et al. (2020) empirically established continuance as an outcome of confirmation.

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Bhattacherjee (2001) defined confirmation as a psychological or affective state related to and resulting from a cognitive appraisal of the expectation-performance discrepancy. The premise of confirmation is characterized when there is an incongruity between pre-use expectations and perceived post-use performance (Oghuma et al., 2016; Veeramootoo et al., 2018). Following Oghuma et al. (2016), confirmation is defined in this study, as a perception of the congruence between a user's initial expectations and the actual performance of an AIVA.

Technological readiness (TR) aims to understand how likely people are to accept and use new technologies. TR is a behavioral attitude towards accepting new technologies irrespective of the technology (Blut & Wang, 2020; Parasuraman & Colby, 2015; Roy et al., 2018). Most studies have identified four dimensions—innovativeness, optimism, discomfort, and insecurity, that collectively explain technology usage (Blut & Wang, 2020; Park & Zhang, 2022). Following Blut and Wang (2020), TR is conceptualized in this study as composed of the motivators (optimism and innovativeness) and the inhibitors (discomfort and insecurity). Individuals with higher TR tend to have more favorable attitudes towards new technologies in terms of quality, value, satisfaction, etc., which results in a higher tendency to use technology or continue using it (Park & Zhang, 2022).

Optimism and innovativeness, as TR constructs, positively influence confirmation of expectations about technology (Blut & Wang, 2020; Park & Zhang, 2022). Optimists tend to have constructive attitudes towards technology, and they see more benefits in specific technologies. Therefore, individuals with high level of optimism towards technology are more likely to fulfil their expectations of AIVAs. In addition, individuals with high levels of innovativeness, who corresponds to greater risk acceptance as they enjoy the novelty of new technologies, also have a positive effect on confirmation (Blut & Wang, 2020). Innovative users will also be more likely to enjoy the usefulness of AIVAs compared to those users with lower levels, especially with recurring use context such as obtaining the right search results. It has been shown that this characteristic of innovativeness also plays a role in use continuance of smart and disruptive technologies (Schmidthuber et al., 2020).

On the other hand, users who experience discomfort with new technology often lack knowledge and confidence in its use, resulting in increased anxiety (Blut & Wang, 2020). Discomfort and insecurity are TR inhibitors that lower an individual's propensity to accept and use new technology (Blut & Wang, 2020). Consequently, individuals with higher levels of discomfort and insecurity often have lower expectations and lower perceived usefulness, leading to negative effects on confirmation. Discomfort is characterized as the sense of oppression by new technology, while insecurity arises from skepticism about the viability of new technology (Parasuraman, 2000). Users with greater discomfort may find specific technologies overwhelming and may also have difficulty with using or reluctance to explore additional features, which may prevent them from receiving the full AIVA experience and lead to reduced levels of confirmation. Therefore, users with higher discomfort and insecurity are expected to have lower levels of confirmation and may not fully realize the benefits of using AIVAs, such as improved efficiency and flexibility. Based on the literature reviewed, the following hypotheses are formulated:

H1a. Optimism has a positive effect on confirmation.

H1b. Innovativeness has a positive effect on confirmation.

H1c. Discomfort has a negative effect on confirmation.

H1d. Insecurity has a negative effect on confirmation.

User's pre-use beliefs and expectations can be disconfirmed after actual use. Many studies have extended this concept of instrumentality to other functional attributes and usage contexts (Lee & Sheehan, 2021). Accordingly, the same disconfirmation may apply to the following four AI-related instrumentalities i.e., Intelligence, Anthropomorphism, Information Quality, and System Quality), which are highly pertinent in the AI-based technology and use context.

Intelligence is often found to be a vital characteristic of AI especially with cognitive and reasoning abilities becoming fundamental architecture of AI algorithms (Hossain et al., 2020). AIVAs are intelligent systems as they autonomously deliver effectual output to a user's requests and continuously learn from the user's behavior. When users have their pre-use expectations confirmed, it indicates that the system is meeting or exceeding their initial assumptions about the AIVA's intelligence. This confirmation is expected to positively reinforce the users' perception of the AI's intelligence, leading to a greater appreciation of its capabilities. Furthermore, as users interact more with the AIVA and experience its intelligent behavior, they become more aware of its ability to adapt and improve over time, which in turn enhances their perception of the AI's intelligence.

On the other hand, anthropomorphism involves the ascription of human-like qualities to non-human agents, leading to an inductive inference based on these features. As noted by Salles et al. (2020), anthropomorphism goes beyond describing existing physical features or behaviors by signifying a human-like interpretation of these features and behaviors. Confirmation of pre-use expectations can have a positive effect on anthropomorphism because users are more likely to ascribe human-like attributes to a system when it meets or exceeds their expectations. As the AIVA provides personalized and contextually relevant responses to user queries, users may perceive the AI as being more human-like in its understanding and communication abilities. Additionally, when users observe the AIVA adapting to their preferences and learning from their interactions, they may further perceive the system as possessing human-like qualities, such as adaptability and learning capabilities. This phenomenon can have a significant impact on user perceptions and experiences, as it enables users to relate to and engage with AI systems more naturally (Salles et al., 2020). According to Epley et al. (2007), two main motivational determinants primarily trigger anthropomorphism. The first is the human need for social connections, which can be extended to non-human agents (Edwards et al., 2019). Secondly, humans need to make

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sense of and interact effectively with the surrounding environment. The ability to judge only with a few visual cues and accurately conclude that an object or human is innately ingrained within humans (Ambady & Rosenthal, 1992). This leads to the following two hypotheses:

H2a. Confirmation has a positive effect on intelligence.

H2b. Confirmation has a positive effect on anthropomorphism.

Wixom and Todd (2005) demonstrated that AI-based services involve information and system characteristics that shape quality perceptions and satisfaction. Information and system qualities are measures of the success of an IS system. Information quality is characterized by the accuracy, currency, presentation, and completeness (reliability) of AI service capabilities. On the other hand, system quality is another facet of AI quality, representing a user's perceived quality of the system's overall performance through flexibility, timeliness, and reliability of the system (Wixom & Todd, 2005).

In the context of AIVAs, users' expectations of information and system qualities will continually evolve as they interact with the AI system. Confirmation of pre-use expectations regarding system quality can lead users to perceive AIVA as more flexible, timely, and reliable in its overall performance. The more users' expectations are confirmed, the greater the perceived levels of AI instrumentality and quality will be. This positive relationship between confirmation and perceived information and system qualities suggests that meeting or exceeding users' expectations is crucial for fostering a sense of satisfaction and continued use of the AIVAs. This understanding leads to the following hypotheses:

H2c. Confirmation has a positive effect on information quality.

H2d. Confirmation has a positive effect on system quality.

AI agents are often perceived to be more competitive than other technologies and users expect them to exhibit higher levels of intelligence (Huang & Rust, 2018). In the context of AIVAs, users' perceptions of intelligence are influenced by the system's language-processing capabilities, proactivity, and autonomy, which in turn generate cues related to trust (Moussawi et al., 2021). Furthermore, users continuously evaluate a brand to determine whether it is fulfilling its performance promises (Dwivedi et al., 2018). Brand trustworthiness reflects a brand's commitment to delivering on its promises, while brand expertise refers to its competence and ability to do so (Erdem & Swait, 2004). When users perceive an AIVA as intelligent, they are more likely to believe that the brand behind the assistant is knowledgeable, experienced, skilled, and qualified and showcasing exceptional engineering. As a result, users' perceptions of AIVA's intelligence can contribute to their beliefs about the brand's trustworthiness. By demonstrating intelligent behavior, AIVAs reinforces users' trust in the brand's ability to deliver on its promises and maintain a high level of performance. Thus, it is highly likely that there is a positive relationship between perceived intelligence and brand trustworthiness. Based on this understanding, the following hypothesis is proposed:

H3a. Intelligence has a positive effect on brand trustworthiness.

Anthropomorphism has a profound influence on users' interaction with AI technologies. It can foster positive attitudes, trust, and engagement among users, which in turn affects their purchase intention and overall satisfaction with the technology Lu et al., 2019; Stroessner & Benitez, 2019; Hsiao & Chen, 2021). This arises because people are naturally inclined to seek social connections and build relationships, even with AI systems (David-Ignatieff et al., 2023; Epley et al., 2007; Pizzi et al., 2023). Research has shown that anthropomorphism can improve users' perceptions of credibility, intelligence, and trust in AI technologies such as chatbots (Cassell & Bickmore, 2003). Similarly, AIVAs with rich anthropomorphic cues can create positive user perceptions and compensate for situations where the message or the technology might have low credibility (Go & Sundar, 2019). By attributing human-like characteristics, AI systems can evoke feelings of warmth, social connection, relatable and engaging connection. This connection may lead users to be more inclined to trust the brand behind the technology (Rauschnabel & Ahuvia, 2014; Seo, 2022; Söderlund & Oikarinen, 2021), believing that the company has a deeper understanding of their needs and desires. Thus, anthropomorphism likely plays a vital role in influencing users' trust in AI technologies and the brands behind them. Therefore, the following hypothesis is proposed:

H3b. Anthropomorphism has a positive effect on brand trustworthiness.

Information and system qualities influence brand trust (Kim & Niehm, 2009; Muhammad et al., 2014; Yuan et al., 2020). The quality of the conversation and the information provided are also necessary measures of intelligence, as proven in various AI-based applications such as live chat agents (Mclean & Osei-Frimpong, 2017) and voice assistants (Yilmaz & Rızvanoğlu, 2021). In the current study, accuracy, currency (up-to-datedness), and completeness (reliability) of information are selected as measures of information quality thus answering users' queries effectively. A technology must produce better levels of user need fulfilment and surpass user expectations to enhance user confidence (Ou & Sia, 2010). Thus, it is hypothesized that:

H3c. Information quality has a positive effect on brand trustworthiness.

The rapid advancements in AIVA technology have had a considerable impact on users' perceptions of system quality and, consequently, their trust in the brand behind the technology. Features such as accent recognition, advanced natural language processing, and comprehension of colloquial speech have greatly improved the user

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experience, ultimately shaping users' emotions and satisfaction with the technology (Nguyen et al., 2022).

High system quality, characterized by reliability, flexibility, and timely responsiveness, contributes to an enhanced brand-user relationship. When users perceive an AIVA system as consistently delivering exceptional performance, adapting to their preferences, and providing seamless user experiences, it signals the brand's commitment to meeting users' needs effectively. This, in turn, strengthens users' trust in the brand (Hasan et al., 2021). By continually improving system quality, brands can enhance user satisfaction, reinforce their image as reliable providers of AI solutions, and foster trust. Based on these insights, the following hypothesis is proposed:

H3d. System quality has a positive effect on brand trustworthiness.

Brand expertise has emerged as a criterion of purchase intention, brand loyalty and success of brand extensions (Featherman et al., 2021). Customers also typically prefer to engage with experts who demonstrate competency and knowledge (Featherman et al., 2021). Thus, AI instrumentality characteristics are fitting proxies of brand expertise. However, there is little discussion on how these AI instrumentality characteristics may impact users' perception of brand expertise of AIVAs. While technology competency could be largely ascribed to system intelligence (Bassellier et al., 2001), it has been established that perceptions of competence contribute to both abstract (namely, spiritual) and concrete (namely, physical, operational) attributes of a brand extends to the products. Furthermore, besides functional attributes, more advanced AIVAs also support user decision-making by reaching a series of conclusions based on input data (Schweitzer et al., 2019). Given the influence of intelligence on the core utility of AIVAs, this leads to the following hypothesis:

H4a. Intelligence has a positive effect on brand expertise.

AIVAs portray a level of humanness with proficient natural language processing and surreal personification of AIVAs. It is a baseline requirement for developers of AIVAs to seek to fuel users' perception as human-like as far as possible to strengthen their connection with them (Habler et al., 2019; Schreibelmayr & Mara, 2022). For example, certain brands endow AIVAs with unique personalities and provide the capability of friendly banter (Steinbruck, 2019). Therefore, leading to the hypothesis:

H4b. Anthropomorphism has a positive effect on brand expertise.

According to Coulter and Coulter (2003), users' beliefs in brand expertise, as represented by engineering and operational excellence, may serve to mitigate concerns related to privacy and technology. Enhancing the user experience through system quality and information quality is a critical factor in developing improved levels of user cognition (Saeed & Abdinnour-Helm, 2008). Insufficient information quality, which can manifest in inaccurate, outdated, or unreliable information, may lead to users expending more time and effort than usual and may prompt them to seek alternative solutions. Conversely, AIVAs that provide users with timely, precise, and relevant information may foster a more favorable perception of the brand's expertise.

Therefore, the following hypothesis is proposed:

H4c. Information quality has a positive effect on brand expertise.

User beliefs in brand expertise, which is typically represented by engineering and operational excellence, can facilitate the reduction of privacy and technology concerns (Coulter & Coulter, 2003). However, for users to develop positive perceptions of brand expertise, AIVAs need to consistently deliver high-quality performance that aligns with users' expectations. System quality, which is one of the key dimensions of service quality, has been shown to play a vital role in shaping users' perceptions of the quality of a service (Parasuraman, 2000). In the context of AIVAs, system quality refers to the extent to which the technology performs its intended functions accurately and efficiently (Saeed & Abdinnour-Helm, 2008). Furthermore, quality has a signaling effect on users' evaluations of brands (Aaker, 1991). AIVA users expect to reap the most benefits with the least possible effort, and consistent and credible performance can help reduce users' cognitive effort in performing personal tasks. If an AIVA consistently delivers high-quality performance, users are likely to associate the brand with expertise, which could lead to more favorable attitudes towards the brand as well reduce the perception of risk (Erdem & Swait, 1998). Therefore, it is reasonable to postulate that high system quality would lead to more positive perceptions of brand expertise. This leads to the following hypothesis:

H4d. System quality has a positive effect on brand expertise.

Brand credibility (based on brand trustworthiness and expertise) is linked to higher perceived value, improved perceptions of the brand's attributes, and lowered perceived risks (Baek et al., 2010). On the other hand, satisfaction is a user's fulfilment response to whether a product, service, or technology feature has offered an agreeable level of consumption-related fulfilment (Oliver, 2014). It exemplifies a general fulfilment belief instead of merely a transactional one. Brand trustworthiness lowers perceived risk by enhancing users' trust concerning the brand's promises. A study on drivers of customer lifetime value proved that brand credibility significantly affected customer satisfaction (Dandis et al., 2022). Sweeney and Swait (2008) found that brand trustworthiness and expertise are continuing summaries of a user's interaction with a brand, which explains satisfaction. Therefore, this leads to the following hypotheses:

H5a. Brand trustworthiness has a positive effect on satisfaction.

H5b. Brand expertise has a positive effect on satisfaction.

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AIVAs are now perceived as collaborators in consumer decision making (Seeber et al., 2020) and could redesign consumer journeys by providing personalized experiences (Yang et al., 2020). Confirmation has positive influence on satisfaction which in turn has an impact on use continuance intention (Bhattacherjee, 2001). Satisfaction affects attitude, beliefs and behaviors towards a technology (Bhattacherjee & Premkumar, 2004) and system success (Delone & McLean, 2014). Many studies have validated the relationship between satisfaction and continuance intention across different contexts (Ambalov, 2018; Lee & Sheehan, 2021). Thus, leading to the last two hypotheses:

H6a. Confirmation has a positive effect on satisfaction.

H6b. Satisfaction has a positive effect on use continuance intention.

Figure 1 presents the proposed conceptual model.

3 | METHODOLOGY: SAMPLING AND DATA COLLECTION

3.1 | Sampling procedure

In line with common practices in use continuance literature, this research employed a nonprobability purposive sampling strategy, targeting individuals who had used AIVAs at least once in the past month. The primary goal was to identify factors influencing AIVA continued usage among existing users, regardless of their expertise. Malaysia was selected as the study context due to its high AIVA usage, with one in three Malaysians using voice search or command features (Farveen, 2020). Notably, younger individuals showed a higher preference for AIVAs, with 46% of Malaysians aged 16–24 using an AIVA (Goh, 2021). A screening question ensured the inclusion of current AIVA users, and participants had to be at least 18 years old in accordance with the university's ethical committee guidelines.

All survey questions were marked as required, preventing respondents from skipping any items. To encourage participation and boost response rates, incentives were offered in the form of meal donations for each completed response (to the Lost Food Project, Kuala Lumpur) and Touch and Go vouchers awarded via a random draw. These incentives were modest enough to avoid introducing bias, while still effectively motivating participants to complete the survey (Wolf et al., 2016).

3.2 | Sample size determination

The minimum required sample size was calculated using G*Power 3.1 software, based on power analysis considerations. With effect size, significance level, and statistical power set at 0.15, 0.05, and 0.95,

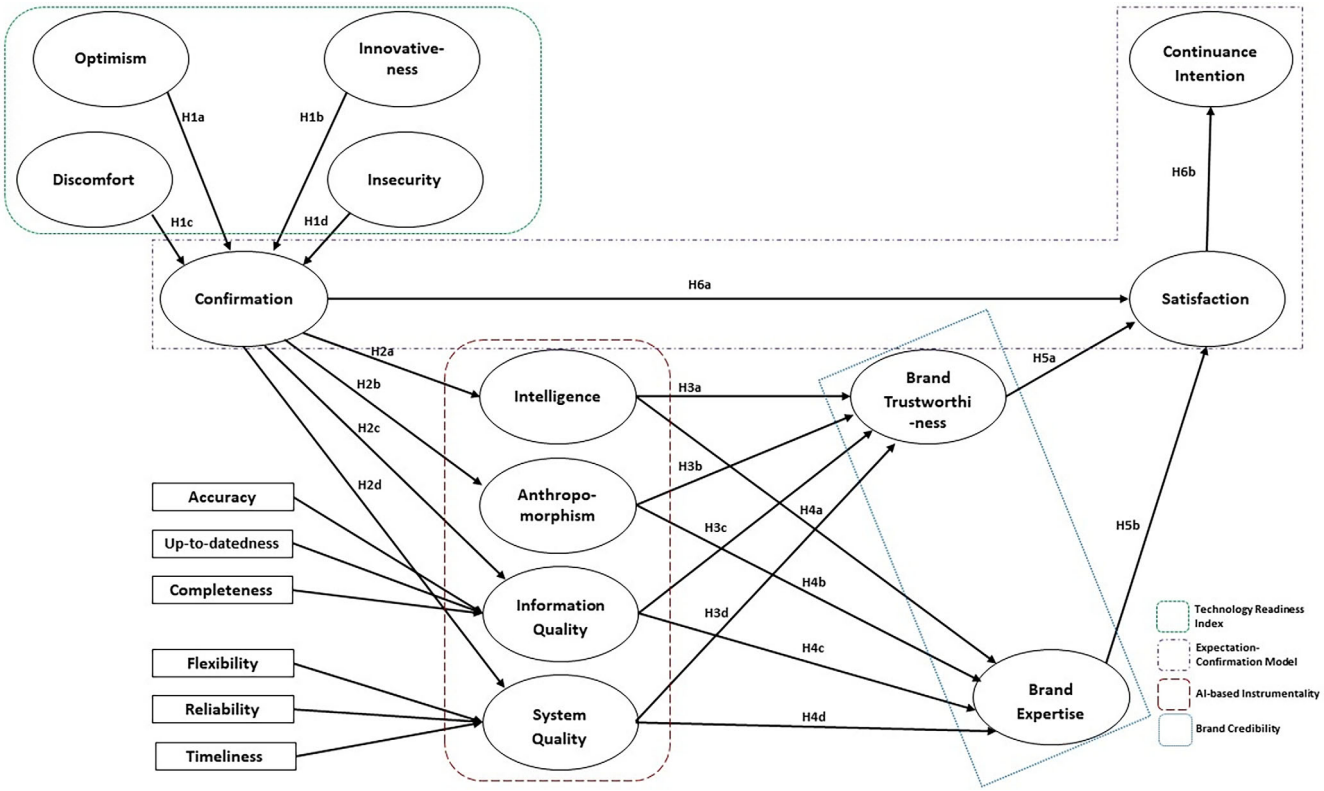


FIGURE 1 Conceptual model.

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respectively, the software indicated a minimum sample size of 208. To circumvent potential issues associated with small sample sizes, the minimum sample size for this study was established at 250.

3.3 | Operationalization of constructs and research instrument

The survey questions were adapted from existing literature to suit the context of this study, focusing on AIVAs. All items were measured using a 7-point scale, ranging from 1 ("strongly disagree") to 7 ("strongly agree"), allowing respondents to indicate the option that best described their experience with their AIVA brand. Questions were customized and adapted from established scales for the AIVA context. The definitions of the constructs, the items as well as the sources used in this study are given in Table A1. A preliminary version of the survey was first tested with academic experts and potential participants. Following this pre-test, adjustments were made (e.g., substituting words and enhancing clarity) to develop the final version of the questionnaire.

3.4 | Measurement model assessment

The partial least square-structural equation modelling (PLS-SEM) approach using Smart PLS v3.3.7 software, was selected, because it provides vigorous means to evaluate survey-based data, due to the complexity of model and suitability when data are not normally distributed. Higher-order constructs (HOCs) reduce path models to improve parsimony and expand construct conceptualizations (Sarstedt et al., 2019). In this study, Information Quality and System Quality are reflective-formative Type II HOC. Both constructs are assigned as 'distinct ideas' and not theoretically consolidated as second-order components with three measurements variables each. For Information Quality, the variables were Accuracy, Completeness and Up-to-datedness, while for System Quality, the variables were Flexibility, Reliability and Timeliness. Using embedded two stage method, the latent scores of all items under the first-order component were used for the HOCs. The remaining constructs in the model were assessed by capturing the latent scores from the previous stage.

4 | RESEARCH RESULTS

A total of 281 responses were validated for data analysis. There were no incidences of straight-lining or outliers. There were slightly more Female respondents (59.1%). Some studies have indicated that women are more likely to participate in surveys than males (e.g., Curtin et al., 2000). As for age, 42% were between 18 and 24; 37.4% between 25 and 34 and 20.6% were 35 years old and above. Most respondents hold a bachelor's degree (57.7%), followed by postgraduate degree (25.3%), and others such as diploma, vocational school, high school or did not complete high school (17.0%). 45.9% of the respondents were full-time students followed by full-time employees (42.7%) and others (11.4%) made up the rest. They included self-employed, part-timers, homemakers and the unemployed.

TABLE 1 Multivariate skewness and kurtosis.

	b	*z*	*p* value
Skewness	63.9260	2993.86771	0
Kurtosis	431.5352	35.79132	0

The respondents were also asked to name the brand most used. It was found that the majority used Google Assistant (48.0%), followed by Siri (43.4%) and others such as Bixby, Cortana, Alexa, Genie and Xiao Ai (8.6%). Experience of using AIVAs ranges from less than a year (27.4%), 1–2 years (24.2%), 2–4 years (21.4%) and more than 4 years (27.0%). Respondents who used AIVAs at least once a day stood at 31.3%, followed by those who used at least once a week (39.5%) and rarely (2–3 times or almost never in a month) (29.2%). Interestingly, 54.4% of the users identified themselves as frequent users of AIVAs. Table A2 shows the descriptive statistics and variability, including central tendency.

The outcomes of the multivariate skewness and kurtosis of the model assessment (Table 1) indicate the non-multivariate normal characteristics of the data and the need for a non-parametric approach in the subsequent analysis.

The measurement model was assessed via a two-step approach. First, the validity and reliability of the instrument were examined. The outer loadings of all the items are above 0.7, except for the items INS1, INS4, DIS1, SYT1, INA2 and BT2. Hence, these items were removed as they were below the threshold of 0.7. Retaining these items in the constructs would lower the internal consistency reliability of the constructs. All the constructs had an AVE value of >0.5, indicating the measurement model has convergent validity. The composite reliability (CR) and Cronbach's alpha (α) (with the six items removed) exceeded the threshold of 0.7, suggesting internal consistency reliability (see Table 2). Next, the structural model was then assessed to test the proposed relationships between constructs.

Since the research has two second-order components, namely, Information Quality and System Quality, the validity and reliability of their measurement items were also evaluated and found to be valid and reliable (Table 3) (Sarstedt et al., 2019). All first-order measurements had significant positive relationships with the second-order constructs. As for System Quality, it had a significant positive relationship with two first-order dimensions. Additionally, the variance inflation factor (VIF) for all dimensions was examined, revealing that collinearity among the three dimensions was below the threshold level of 3.3 (Hair et al., 2017). This indicates that there was no significant multicollinearity among the three dimensions of Information Quality and System Quality respectively, and the coefficients' effects can be interpreted without issues.

Discriminant validity was assessed using the Heterotrait-Monotrait Ratio of Correlations (HTMT) method (Henseler et al., 2015). Most HTMT results fell below the conservative threshold

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TABLE 2 Convergent validity assessments.

Constructs	Items	Outer loading	α	CR	AVE
Innovativeness	INN1	0.735	0.822	0.881	0.649
	INN2	0.803			
	INN3	0.830			
	INN4	0.850			
	INN1	0.735			
Optimism	OPT1	0.852	0.841	0.893	0.677
	OPT2	0.834			
	OPT3	0.805			
	OPT4	0.798			
Insecurity	INS2	0.698	0.694	0.834	0.721
	INS3	0.978			
Discomfort	DIS2	0.736	0.760	0.860	0.673
	DIS3	0.862			
	DIS4	0.840			
Confirmation	CON1	0.943	0.941	0.962	0.894
	CON2	0.946			
	CON3	0.949			
Satisfaction	SAS1	0.928	0.939	0.956	0.845
	SAS2	0.927			
	SAS3	0.942			
	SAS4	0.880			
Continuance intention	INT1	0.900	0.899	0.93	0.768
	INT2	0.894			
	INT3	0.876			
	INT4	0.833			
Intelligence	PI1	0.806	0.920	0.937	0.714
	PI2	0.840			
	PI3	0.840			
	PI4	0.851			
	PI5	0.859			
	PI6	0.873			
Anthropomorphism	PA1	0.795	0.905	0.93	0.726
	PA2	0.894			
	PA3	0.895			
	PA4	0.831			
	PA5	0.841			
Info accuracy	INA1	0.835	0.805	0.911	0.837
	INA3	0.887			
Info completeness (reliability)	INR1	0.919	0.904	0.94	0.839
	INR2	0.927			
	INR3	0.902			
Info Up-to-datedness (recency)	INU1	0.923	0.908	0.942	0.844
	INU2	0.925			
	INU3	0.908			
System flexibility	SYF1	0.930	0.939	0.961	0.892
	SYF2	0.954			

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TABLE 2 (Continued)

Constructs	Items	Outer loading	α	CR	AVE
	SYF3	0.949			
System reliability	SYR1	0.936	0.919	0.948	0.860
	SYR2	0.932			
	SYR3	0.914			
System timeliness	SYT2	0.925	0.856	0.933	0.874
	SYT3	0.917			
Brand trustworthiness	BT1	0.730	0.872	0.907	0.662
	BT3	0.875			
	BT4	0.816			
	BT5	0.842			
	BT6	0.779			
Brand expertise	BE1	0.902	0.952	0.961	0.805
	BE2	0.875			
	BE3	0.888			
	BE4	0.864			
	BE5	0.917			
	BE6	0.937			

Abbreviations: AVE, average variance extracted; CR, composite reliability; α, Cronbach's alpha.

TABLE 3 Assessment of formative constructs.

Second order component	First order component	Outer weights	*p* value	Outer loadings	*p* value	VIF
Information quality	Information accuracy	0.256	.007	0.838	.000	2.156
	Information up-to-datedness (recency)	0.601	.000	0.949	.000	2.228
	Information completeness (reliability)	0.264	.007	0.816	.000	1.948
System quality	System flexibility	0.532	.000	0.900	.000	1.719
	System reliability	0.550	.000	0.913	.000	2.114
	System timeliness	0.036	.295	0.520	.000	1.407

Abbreviation: VIF, variance inflation factor.

of 0.75 (Voorhees et al., 2016), indicating satisfactory discriminant validity. Item loadings were examined, and no instances of cross-stacking with higher values were observed (Table 4). A bootstrapping method with 5000 subsamples produced 95% bias-corrected and accelerated confidence intervals, with none containing the value 1, thus confirming valid and reliable measurement items.

Common method bias was assessed using the 'Grocery Shopping Behavior' marker variable. The minimal difference in R^2 values before (0.489) and after (0.493) adding the marker variable to endogenous constructs suggests that common method bias is not a concern. Additionally, the VIF results in Table 6, ranging from 1.0 to 2.8, are all below the critical threshold of 3.3, indicating no multicollinearity issues.

The structural model was evaluated using the bootstrapping method with 5000 resamples. R^2 value measure each construct's explained variance. Higher R^2 values signify greater explanatory power, with values of 0.75, 0.50, and 0.25 considered significant, moderate, and low, respectively (Hair et al., 2017). The findings show that the endogenous construct, satisfaction, accounts for 48.9% of the variance in continuance intention (See Table 5). Likewise, endogenous constructs, including confirmation, brand expertise, and brand trustworthiness, explain 64.5% of the variance in satisfaction. Information quality, system quality, intelligence, and anthropomorphism account for 39.5% and 35.0% of the variance in brand trustworthiness and brand expertise, respectively. Confirmation, as an endogenous construct, accounts for 48.4%, 61.5%, 48.4%, and 28.6% of the variance in information quality, system quality, intelligence, and anthropomorphism, respectively. All exogenous constructs have Q^2 statistics greater than 0, demonstrating predictive relevance. Six relationships exhibit large effect sizes (f^2 value of 0.35), one with a medium effect (f^2 value of 0.02), eight with small effects (f^2 value of 0.15), and five with no effects (f^2 value below 0.02). Q^2 values were found to be greater than 0 and within the range of 0.19–0.54, suggesting the constructs provide strong support for the model's predictive relevance.

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TABLE 4 Heterotrait-monotrait ratio of correlations.

Constructs		1	2	3	4	5	6	7	8	9	10
1	Anthropomorphism										
2	Brand Expertise	0.546 *[0.465, 0.622]*									
3	Brand Trustworthiness	0.577 *[0.482, 0.661]*	0.731 *[0.634, 0.817]*								
4	Confirmation	0.576 *[0.491, 0.654]*	0.564 *[0.454, 0.670]*	0.577 *[0.455, 0.682]*							
5	Continuance Intention	0.546 *[0.454, 0.633]*	0.577 *[0.464, 0.690]*	0.656 *[0.542, 0.751]*	0.787 *[0.709, 0.857]*						
6	Discomfort	0.163 *[0.119, 0.247]*	0.126 *[0.076, 0.239]*	0.197 *[0.123, 0.311]*	0.226 *[0.100, 0.360]*	0.287 *[0.172, 0.413]*					
7	Innovativeness	0.126 *[0.081, 0.230]*	0.360 *[0.263, 0.453]*	0.333 *[0.239, 0.427]*	0.308 *[0.201, 0.413]*	0.293 *[0.184, 0.405]*	0.091 *[0.079, 0.195]*				
8	Insecurity	0.102 *[0.067, 0.205]*	0.060 *[0.054, 0.162]*	0.075 *[0.077, 0.180]*	0.116 *[0.056, 0.240]*	0.078 *[0.053, 0.203]*	0.417 *[0.279, 0.555]*	0.247 *[0.130, 0.380]*			
9	Intelligence	0.614 *[0.528, 0.691]*	0.523 *[0.414, 0.629]*	0.540 *[0.427, 0.642]*	0.747 *[0.652, 0.825]*	0.672 *[0.574, 0.761]*	0.251 *[0.128, 0.390]*	0.308 *[0.212, 0.411]*	0.138 *[0.082, 0.262]*		
10	Optimism	0.410 *[0.303, 0.516]*	0.533 *[0.433, 0.629]*	0.516 *[0.403, 0.622]*	0.451 *[0.335, 0.560]*	0.484 *[0.368, 0.594]*	0.127 *[0.105, 0.228]*	0.250 *[0.160, 0.355]*	0.111 *[0.093, 0.211]*	0.362 *[0.241, 0.479]*	
11	Satisfaction	0.525 *[0.439, 0.604]*	0.661 *[0.566, 0.748]*	0.662 *[0.576, 0.741]*	0.794 *[0.727, 0.856]*	0.758 *[0.678, 0.834]*	0.141 *[0.082, 0.257]*	0.278 *[0.174, 0.383]*	0.048 *[0.037, 0.165]*	0.608 *[0.505, 0.699]*	0.465 *[0.352, 0.568]*

Note: Italicized numbers in brackets represent the 95% bias-corrected and accelerated confidence intervals derived from bootstrapping with 5000 samples.

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TABLE 5 R^2, Q^2 statistics, VIF and effect size.

Endogenous construct	R^2	Q^2	Path	f^2	Effect size	VIF
Confirmation	0.230	0.191	Optimism → Confirmation	0.15	Medium	1.060
			Innovativeness → Confirmation	0.04	Small	1.082
			Discomfort → Confirmation	0.03	Small	1.089
			Insecurity → Confirmation	0.00	No effect	1.113
Intelligence	0.484	0.341	Confirmation → Intelligence	0.94	Large	1.000
Anthropomorphism	0.286	0.203	Confirmation → Anthropomorphism	0.40	Large	1.000
Info Quality	0.484	0.374	Confirmation → Info quality	0.94	Large	1.000
System Quality	0.615	0.406	Confirmation → System quality	1.60	Large	1.000
Brand Expertise	0.350	0.275	Anthropomorphism → Brand expertise	0.08	Small	1.662
			Info quality → Brand expertise	0.02	Small	2.815
			Intelligence → Brand expertise	0.01	No effect	2.595
			System Quality → Brand expertise	0.01	No effect	2.720
Brand Trustworthiness	0.395	0.251	System Quality → Brand trustworthiness	0.08	Small	2.720
			Anthropomorphism → Brand trustworthiness	0.07	Small	1.662
			Intelligence → Brand trustworthiness	0.00	No effect	2.595
			Info Quality → Brand trustworthiness	0.00	No effect	2.815
Satisfaction	0.645	0.539	Confirmation → Satisfaction	0.54	Large	1.507
			Brand Expertise → Satisfaction	0.07	Small	2.006
			Brand Trustworthiness → Satisfaction	0.04	Small	1.981
Continuance Intention	0.489	0.367	Satisfaction → Continuance intention	0.96	Large	1.000

Abbreviations: f^2, effect size; R^2, coefficient of determination; VIF, variance inflation factor.

TABLE 6 Summary of research results.

Hypothesis	Path	Std Beta	Std Dev	*t* value	*p* value	BCI LL	BCI UL	Supported
H1a	Optimism → Confirmation	0.352	0.064	5.527	<.001	0.243	0.453	Yes
H1b	Innovativeness → Confirmation	0.191	0.057	3.331	<.001	0.095	0.283	Yes
H1c	Discomfort → Confirmation	−0.147	0.062	2.368	.009	−0.236	−0.035	Yes
H1d	Insecurity → Confirmation	−0.044	0.065	0.674	.250	−0.116	0.140	No
H2a	Confirmation → Intelligence	0.695	0.050	14.047	<.001	0.595	0.762	Yes
H2b	Confirmation → Anthropomorphism	0.535	0.045	11.873	<.001	0.452	0.602	Yes
H2c	Confirmation → Info Quality	0.696	0.038	18.282	<0.001	0.624	0.75	Yes
H2d	Confirmation → System Quality	0.784	0.029	27.308	<0.001	0.734	0.828	Yes
H3a	Intelligence → Brand Trustworthiness	0.077	0.083	0.928	.177	−0.045	0.235	No
H3b	Anthropomorphism → Brand trustworthiness	0.267	0.068	3.910	<0.001	0.153	0.377	Yes
H3c	Info Quality → Brand Trustworthiness	0.015	0.091	0.160	.437	−0.143	0.154	No
H3d	System Quality → Brand Trustworthiness	0.360	0.109	3.293	<0.001	0.175	0.534	Yes
H4a	Intelligence → Brand Expertise	0.123	0.096	1.284	.100	−0.019	0.296	No
H4b	Anthropomorphism → Brand expertise	0.290	0.063	4.596	<.001	0.188	0.395	Yes
H4c	Info quality → Brand expertise	0.177	0.098	1.798	.036	0.007	0.331	Yes
H4d	System quality → Brand expertise	0.109	0.110	0.993	.160	−0.081	0.277	No
H5a	Brand trustworthiness → Satisfaction	0.220	0.065	3.371	<.001	0.127	0.339	Yes
H5b	Brand expertise → Satisfaction	0.174	0.065	2.673	.004	0.073	0.288	Yes
H6a	Confirmation → Satisfaction	0.538	0.061	8.778	<.001	0.436	0.638	Yes
H6b	Satisfaction → Continuance intention	0.699	0.045	15.576	<0.001	0.616	0.764	Yes

Abbreviations: BCI LL, bias corrected confidence interval lower limit; BCI UL, bias corrected confidence interval lower limit; Std Beta, standardized beta; Std Dev, standard deviation.

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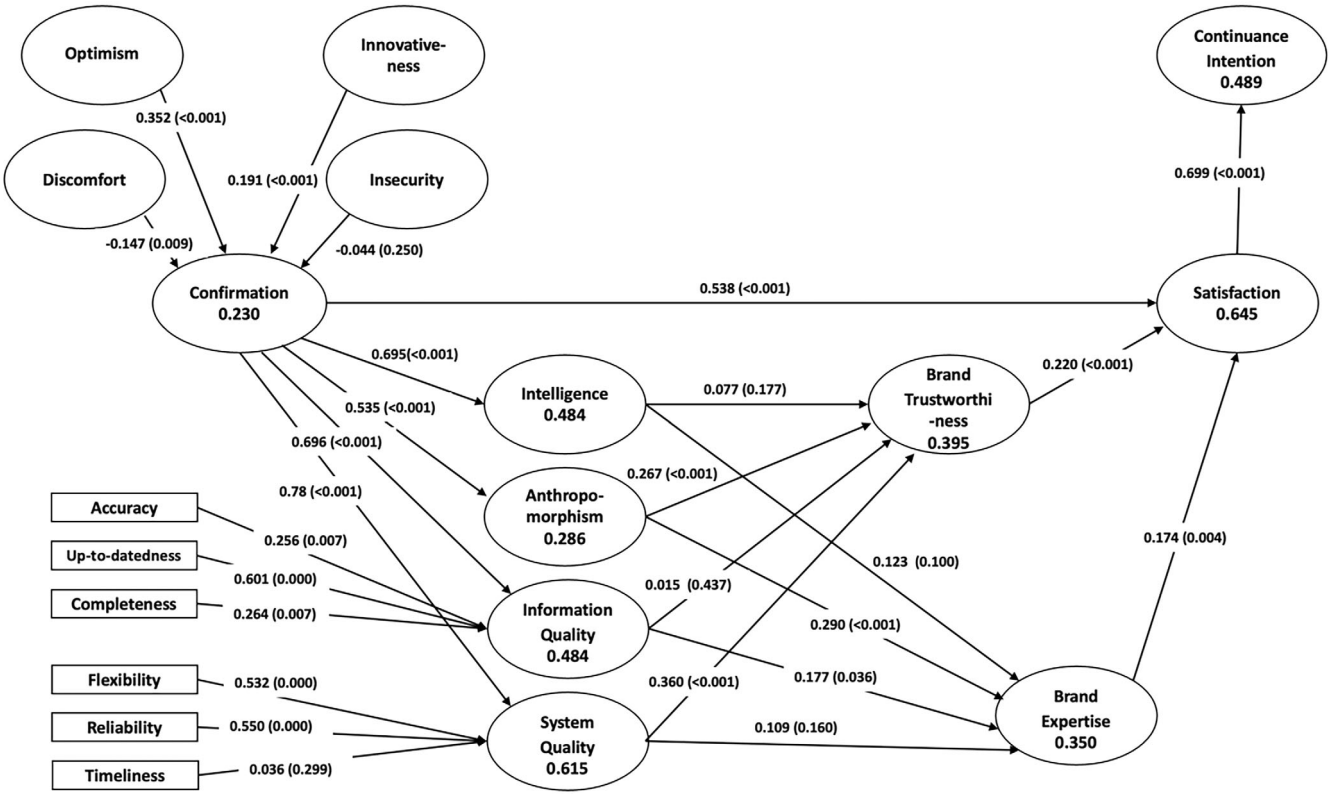


FIGURE 2 Results of the hypotheses testing.

Table 6 summarizes the results of hypothesis testing, and Figure 2 displays the structural model, indicating that most hypotheses are supported.

5 | DISCUSSION

The findings suggest that a user's technology-related personal tendencies play a crucial role in the post-adoption phase of AIVAs. Users with higher levels of optimism, Innovativeness, and lower levels of discomfort are more likely to have their expectations confirmed after using the technology, which subsequently enhances their perception of AI-based instrumentality factors, such as anthropomorphism, intelligence, information quality, and system quality.

The non-significant effect between insecurity and confirmation may be attributed to most respondents being younger and more tech-savvy (79.4% being 34 years old and younger). Younger and better-educated individuals tend to be less skeptical of new technologies (Dutot, 2014) and harbor fewer concerns about potentially harmful consequences.

The results also suggest that post-adoption confirmation has a positive impact on the perceived instrumentality factors of AIVAs. When users find that their initial expectations align with the actual performance of AIVAs, their perceptions of the system's anthropomorphism, intelligence, information quality, and system quality improve, leading to reduced risk perceptions. Interestingly, the path coefficients between confirmation and these four instrumentality variables are observed to be higher than that of satisfaction, indicating that post-adoption confirmation has a stronger association with users' perception of AI instrumentality than satisfaction alone.

The results indicate that not all AI instrumentality attributes are equally important in shaping users' perceptions of AIVA's brand credibility. The findings suggest that among the tested attributes, perceived anthropomorphism acts as a key driver for both brand trustworthiness and expertise. Users are more likely to perceive an AIVA as credible if it exhibits human-like qualities, as this fosters a sense of connection and familiarity. The anthropomorphic qualities may facilitate a more engaging and relatable experience for users, enhancing the overall perception of the brand. In accordance with Chen and Park (2021), anthropomorphism fosters social and task attraction, which in turn strengthens cognitive and emotional trust among customers of a system.

On the other hand, information quality is found to be a driver for brand expertise but not brand trustworthiness. This implies that while accurate, complete, and up-to-date information enhances users' perceptions of a brand's expertise, it may not directly impact their trust in the brand. This could be due to users viewing high information quality as a fundamental requirement for AIVAs, rather than a differentiating factor that contributes to trustworthiness. In the absence of visual cues (e.g., communicative gestures and facial expressions), the accuracy, completeness (reliability), and timeliness (recency) of information delivered by AIVAs become even more critical in shaping perceptions

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of their competence. High information quality tends to satisfy customers' needs, fostering a positive perception of a brand's expertise or competency.

Additionally, given the lack of anthropomorphic physical form of AIVAs, the quality of conversations and brand expertise perceptions may be indirectly influenced by information quality. This highlights the importance of providing accurate and relevant information to enhance users' perceptions of AIVA's expertise.

In contrast, system quality is found to be a driver for brand trustworthiness but not brand expertise. A user-friendly, reliable system can enhance users' trust in AIVAs, as it demonstrates the brand's ability to deliver a seamless experience. However, it may not directly contribute to brand expertise perceptions, as users might associate system quality with technical expertise rather than domain-specific knowledge or skills. AI adoption primarily manifests through efficient technical system infrastructure (Delone & McLean, 2014). As 'faceless' services, AIVAs' system quality attributes become more essential in influencing trusting beliefs, i.e., the more user-friendly and reliable a system, the more credible and trustworthy a brand appears to customers (Gao & Waechter, 2017).

If customers frequently encounter issues (e.g., slow responses, service interruptions) or difficult-to-use systems, it may negatively impact users' evaluations of an AIVA brand's ability, increasing risk perceptions and reducing trustworthiness perceptions (Liu et al., 2021). Consequently, brands must prioritize providing a seamless and reliable user experience to foster trust and loyalty among their customers. This includes addressing any technical issues that may arise and continually refining the system to ensure user satisfaction and maintain a positive brand image.

Interestingly, intelligence is not found to be a driver of either brand expertise or brand trustworthiness. This implies that while intelligence is a valuable attribute, users may prioritize other factors, such as anthropomorphism, system quality, and information quality, when assessing an AIVA's brand credibility. Once an AI agent achieves a sufficient level of intelligence, further improvements do not significantly impact consumers' perception of the brand's expertise and trustworthiness. As an AI agent becomes more human-like, users generally form positive perceptions, albeit only up to a certain point. When AIVAs fail to meet expectations, users may experience disappointment, negatively affecting the relationship between perceived intelligence and brand expertise or trustworthiness (Abrams, 2021). Another possible explanation is that systems focusing on establishing emotional connections and understanding user needs may be perceived as more trustworthy and demonstrate expertise, even if their intelligence is not significantly higher than other systems. Additionally, highly intelligent systems may be complex and challenging for users to comprehend. Users might prefer more accessible and user-friendly systems, even if they are less intelligent. In this case, perceived intelligence, although essential, might not positively contribute to brand expertise and trustworthiness.

Brand credibility, encompassing both expertise and trustworthiness, is a key determinant of user satisfaction and continued use or loyalty (Sweeney & Swait, 2008). Credible brands are perceived to deliver high-quality products, leading to increased user satisfaction. As satisfaction grows, users are more likely to develop loyalty towards the brand, further reinforcing brand credibility (Lee et al., 2021; Sweeney & Swait, 2008). Our study reinforces the significance of these connections in the context of AIVAs. The direct effects of brand expertise and brand trustworthiness on post-use satisfaction are found to be significantly positive. Beliefs that a brand is an expert and trustworthy lends credibility to its products, indirectly fostering use continuance intention through repeated satisfaction experiences (Maroufkhani et al., 2022). Consequently, it is crucial for AIVA brands to focus on key drivers of brand credibility, such as anthropomorphism, system quality, and information quality, to enhance user satisfaction, loyalty, and overall success in the market.

5.1 | Theoretical implications

The study extends the theoretical application of the ECM in the context of AIVAs by integrating AI instrumentalities and brand credibility in shaping user satisfaction and use continuance intention. The study demonstrates that confirmation of user expectations plays a crucial role in shaping satisfaction and, ultimately, continuance intention. The findings on the impact of confirmation on AI instrumentality attributes highlight the importance of confirming users' initial expectations as soon as possible, as this can have a direct impact on satisfaction and the perceived instrumentality of AI agents. Overall, the study provides a more comprehensive understanding of the factors that influence the adoption and usage of AIVAs, offering a richer perspective on the dynamics of this emerging technology.

The study also contributes to the literature on brand credibility in the AI domain by identifying the drivers of brand expertise and trustworthiness, with information quality, perceived anthropomorphism, and system quality emerging as key drivers. In this context, brands should prioritize delivering high-quality information to satisfy users' needs and expectations, ultimately strengthening their perceived expertise in the market. The insights derived from this study can inform the strategies of product developers, service providers, and marketers in addressing the quality dimensions of AIVAs and improving user confirmation, satisfaction, and use continuance. It is crucial for brands to implement strategies that confirm users' initial expectations as soon as possible and appeal to the optimist and innovative side of users to ensure stickiness of technology.

5.2 | Managerial implications

From a practical perspective, marketers should consider providing proper support channels and materials to alleviate any discomfort that may arise during usage. Developers should also focus on improving reliability and intelligence, such as the ability to understand different languages and accents through advanced NLP and deep learning programs, to improve perceived instrumentality performance. Overall, this study provides a comprehensive understanding of user

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expectations and confirmations, as well as the subsequent impact on perceived instrumentality and satisfaction, and offers a richer perspective on the dynamics of AIVA adoption and usage.

5.3 | Limitations and future research

The present study has some limitations that future research should address to enhance the robustness and generalizability of findings. One limitation is the exclusive reliance on quantitative methods, which may neglect the intricacies in user behavior and attitudes. Future research should incorporate qualitative approaches, such as interviews and focus groups, to deepen the understanding of the identified relationships.

Moreover, the study focused on only four AI-related instrumentality factors. Future research should explore additional attributes such as privacy, trust, service quality, and device design, to obtain a more comprehensive understanding of post-use behavior. Furthermore, future studies could investigate the impact of context and use cases on AIVA adoption and continued use, and explore the applicability of other theoretical models, such as the Diffusion of Innovations or the Task-Technology Fit Model, to better understand the factors driving AIVA adoption and continued use.

Additionally, future research should include participants from diverse countries and conduct cross-cultural comparisons to increase global relevance. This comparative analysis will help identify potential cultural influences on user behavior and preferences, ultimately enriching the understanding of AIVA adoption and usage globally.

Future research should further explore the role of perceived intelligence and its relationship with brand expertise and trustworthiness, as well as the potential influence of other factors, such as emotional connection and user accessibility. Additionally, researchers could investigate the potential impact of demographic factors, such as age and education, on user perceptions of AIVAs, as these factors may contribute to differences in skepticism, expectations, and experiences.

6 | CONCLUSION

In conclusion, this study extends the ECM to the context of AIVAs and demonstrates the significance of technology-related personal tendencies in shaping post-adoption attitudes towards AIVAs. Optimism, Innovativeness, and Discomfort are important traits in determining the perceived level of intelligence, anthropomorphism, information quality, and system quality of AIVAs. Moreover, post-use confirmation is found to have a stronger association with the perceived instrumentality of AIVAs than post-use satisfaction. The study also reveals that perceived anthropomorphism and information quality are key drivers of brand expertise, which links to satisfaction and use continuance intention. On the other hand, perceived anthropomorphism and system quality are significant factors in shaping brand trustworthiness. Overall, the study highlights the importance of building brand credibility through providing positive post-adoption experiences to foster satisfaction, loyalty, and continued use.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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REFERENCES

Aaker, D. (1991). *Managing brand equity*. Free Press.
Abrams, Z. (2021). *The promise and challenges of AI*. American Psychological Association.
Ambady, N., & Rosenthal, R. (1992). Thin slices of expressive behavior as predictors of interpersonal consequences: A meta-analysis. *Psychological Bulletin*, *111*(2), 256–274.
Ambalov, I. (2018). A meta-analysis of IS continuance: An evaluation of the expectation-confirmation model. *Telematics and Informatics*, *35*(6), 1561–1571.
Atieh, P. (2021). Humanizing voice assistant: The impact of voice assistant personality on consumers' attitudes and behaviors. *Journal of Retailing and Consumer Services*, *58*, 102283. https://doi.org/10.1016/j.jretconser.2020.102283
Baek, T., Kim, J., & Yu, J. (2010). The differential roles of brand credibility and brand prestige in. *Psychology & Marketing*, *27*(7), 662–678.
Bassellier, G., Reich, B., & Benbasat, I. (2001). Information technology competence of business managers: A definition and research model. *Journal of Management Information Systems*, *17*(4), 159–182.
Bem, D. J. (1972). Self-perception theory. *Advances in Experimental Social Psychology*, *6*, 1–62.
Bhattacherjee, A. (2001). Understanding information systems continuance: An expectation-confirmation model. *MIS Quarterly*, *25*(3), 351–370.
Bhattacherjee, A., & Premkumar, G. (2004). Understanding changes in belief and attitude toward information technology usage: A theoretical model and longitudinal test. *MIS Quarterly*, *28*(2), 229–254.
Blut, M., & Wang, C. (2020). Technology readiness: A meta-analysis of conceptualizations of the construct and its impact on technology usage. *Journal of the Academy of Marketing Science*, *48*, 649–669.
Cassell, J., & Bickmore, T. (2003). Negotiated collusion: Modeling social language and its relationship. *User Modeling and User-Adapted Interaction*, *13*, 89–132.
Chen, Q., & Park, H. (2021). How anthropomorphism affects trust in intelligent personal assistants. *Industrial Management & Data Systems*, *12*, 2722–2737.
Coulter, K., & Coulter, R. (2003). The effects of industry knowledge on the development of trust in service relationships. *International Journal of Research in Marketing*, *20*, 31–43.

14791838, 2024, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cb.2228, Wiley Online Library on [23/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Curtin, R., Presser, S., & Singer, E. (2000). The effects of response rate changes on the index of consumer sentiment. *Public Opinion Quarterly*, *64*, 413–428.

Dandis, A., Al Haj Eid, M., Robin, R., & Wierdak, N. (2022). An empirical investigation of the factors affecting customer lifetime value. *International Journal of Quality & Reliability Management*, *39*, 910–935.

David-Ignatieff, A., Buzeta, C., De Pelsmacker, P., & Ben Dahmane Mouelhi, N. (2023). This embodied conversational agent looks very human and as old as I feel! The effect of perceived agent anthropomorphism and consumer-agent age difference on brand attitude. *Journal of Marketing Communications*. https://doi.org/10.1080/13527266.2023.2199026

Delone, W., & McLean, E. (2014). The DeLone and McLean model of information systems success: A ten-year update. *Journal of Management Information Systems*, *19*(4), 9–30.

Dutot, V. (2014). Adoption of social media using technology acceptance model: The generational effect. *International Journal of Technology and Human Interaction*, *10*(4), 18–35.

Dwivedi, A., Nayeem, T., & Murshed, F. (2018). Brand experience and consumers' willingness-to-pay (WTP) a price premium mediating role of brand credibility and perceived uniqueness. *Journal of Retailing and Consumer Services*, *44*, 100–107.

Edwards, C., Edwards, A., Stoll, B., Lin, X., & Massey, N. (2019). Evaluations of an artificial intelligence instructor's voice: Social identity theory in human-robot interactions. *Computers in Human Behavior*, *90*, 357–362.

Epley, N., Waytz, A., & Cacioppo, J. (2007). On seeing human: A three-factor theory of anthropomorphism. *Psychological Review*, *114*(4), 864–886.

Erdem, T., & Swait, J. (1998). Brand equity as a signaling phenomenon. *Journal of Consumer Psychology*, *7*(2), 131–157.

Erdem, T., & Swait, J. (2004). Brand credibility, brand consideration, and choice. *Journal of Consumer Research*, *31*(1), 191–198.

Farveen, F. (2020). 5 myths about voice marketing among marketers in Malaysia. Advertising+marketing. https://www.marketing-interactive.com/5-myths-about-voice-marketing-among-marketers-in-malaysia

Featherman, M., Jia, S., Califf, C., & Hajli, N. (2021). The impact of new technologies on consumers beliefs: Reducing the perceived risks of electric vehicle adoption. *Technological Forecasting and Social Change*, *169*, 120847.

Gao, L., & Waechter, K. (2017). Examining the role of initial trust in user adoption of mobile payment services: An empirical investigation. *Information Systems Frontiers*, *19*, 525–548.

Go, E., & Sundar, S. (2019). Humanizing chatbots: The effects of visual, identity and conversational cues on humanness perceptions. *Computers in Human Behavior*, *97*, 304–316.

Goh, V. (2021). *Here comes voice. IAB sea+india*. IAB Southeast Asia and India. https://www.iabseaindia.com/blog/voice-is-coming-to-southeast-asia-and-india-whether-brands-are-ready-or-not

Habler, F., Schwind, V., & Henze, N. (2019). *Effects of smart virtual Assistants' gender and language. Proceedings of mensch und computer 2019 (MuC19), 469–473*. ACM.

Hair Jr, J. F., Hult, G. T. M., Ringle, C. M., & Sarstedt, M. (2021). *A primer on partial least squares structural equation modeling (PLS-SEM)* (2nd ed.). Sage Publications.

Hasan, R., Shams, R., & Rahman, M. (2021). Consumer trust and perceived risk for voice-controlled artificial intelligence: The case of Siri. *Journal of Business Research*, *131*, 591–597.

Henseler, J., Ringle, C., & Sarstedt, M. (2015). New criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, *43*(1), 115–135.

Hossain, M., Akter, S., & Yanamandram, V. (2020). Revisiting customer analytics capability for data driven retailing. *Journal of Retailing and Consumer Services*, *56*, 102187.

Hsiao, K., & Chen, C. (2021). What drives continuance intention to use a food-ordering chatbot? An examination of trust and satisfaction. *Library Hi Tech.*, *40*(4), 929–946.

Huang, M., & Rust, R. (2018). Artificial Intelligence in Service. *Journal of Service Research*, *21*(2), 155–172.

Jain, S., Basu, S., Dwivedi, Y. K., & Kaur, S. (2022). Interactive voice assistants – Does brand credibility assuage privacy risks? *Journal of Business Research*, *139*, 701–717.

Kim, H., & Niehm, L. (2009). The impact of website quality on information quality, value, and loyalty intentions in apparel retailing. *Journal of Interactive Marketing*, *23*, 221–233.

Lee, K., & Sheehan, L. (2021). The continuation and recommendation intention of artificial intelligence-based voice: The influence of personal traits. *Internet Research*, *31*(5), 1899–1939.

Lee, K. Y., Sheehan, L., Lee, K., & Chang, Y. (2021). The continuation and recommendation intention of artificial intelligence-based voice assistant systems (AIVAS): The influence of personal traits. *Internet Research*, *31*(5), 1899–1939.

Lee, M., & Park, J. S. (2022). Do parasocial relationships and the quality of communication with AI shopping chatbots determine middle-aged women consumers' continuance usage intentions? *Journal of Consumer Behaviour*, *21*(4), 842–854.

Lim, W. M., Kumar, S., Pandey, N., Verma, D., & Kumar, D. (2023). Evolution and trends in consumer behavior: Insights from journal of consumer behaviour. *Journal of Consumer Behaviour*, *22*(1), 217–232.

Liu, Y., Gan, Y., Song, Y., & Liu, J. (2021). What influences the perceived trust of a voice-enabled smart home system: An empirical study. *Sensors*, *21*, 2037.

Lu, L., Cai, R., & Gursoy, D. (2019). Developing and validating a service robot integration willingness scale. *International Journal of Hospitality Management*, *80*, 36–51.

Mamun, M. R., Senn, W. D., Peak, D. A., Prybutok, V. R., & Torres, R. A. (2020). Emotional satisfaction and IS continuance behavior: Reshaping the expectation-confirmation model. *International Journal of Human Computer Interaction*, *36*(15), 1437–1446.

Market Research Future. (2021). Market Research Future. Voice Assistant Market Size USD 7.30 Billion by 2025, Registering a 24.32% CAGR. https://www.marketsandmarkets.com/Market-Reports/conversational-ai-market-49043506.html

Maroufkhani, P., Asadi, S., Ghobakhloo, M., Jannesari, M. T., & Ismail, W. K. W. (2022). How do interactive voice assistants build brands' loyalty? *Technological Forecasting and Social Change*, *183*, 121870.

Mclean, G., & Osei-Frimpong, K. (2017). Examining satisfaction with the experience during a live chat service encounter-implications for website providers. *Computers in Human Behavior*, *76*, 494–508.

McLean, G., & Osei-Frimpong, K. (2019). Hey Alexa … examine the variables influencing the use of artificial intelligent in-home voice assistants. *Computers in Human Behavior*, *99*, 28–37.

Meidani, Z., Mohammadian, H., Mobaraki, F., Swoboda, W., Moosavi, G. A., & Zare, S. (2022). Exploring the influence of users' characteristics in continuance intention to use electronic medical records (EMR): Extending the expectation confirmation model (ECM). *Informatics and Technology in Clinical Care and Public Health*, *289*, 430–433.

Moriuchi, E. (2019). Okay, Google! An empirical study on voice assistants on consumer engagement and loyalty. *Psychology and Marketing*, *36*(5), 489–501.

Moussawi, S., Koufaris, M., & Benbunan-Fich, R. (2021). How perceptions of intelligence and anthropomorphism affect adoption of personal intelligent agents. *Electronic Markets*, *31*(2), 343–364.

Muhammad, Z., Yi, F., Naz, A., & Muhammad, K. (2014). An empirical study on exploring relationship among information quality, e-satisfaction, e-trust and young generation's commitment to Chinese online retailing. *Journal of Competitiveness*, *6*(4), 3–18.

Mutum, D. S., Ghazali, E. M., Nguyen, B., & Arnott, D. (2014). Online loyalty and its interaction with switching barriers. *Journal of Retailing and Consumer Services*, *21*(6), 942–949.

14791838, 2024, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cb.2228, Wiley Online Library on [23/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Nguyen, T.-M., Quach, S., & Thaichon, P. (2022). The effect of AI quality on customer experience and brand relationship. *Journal of Consumer Behaviour*, *21*(3), 481–493.

Oghuma, A. P., Libaque-Saenz, C. F., Wong, S. F., & Chang, Y. (2016). An expectation-confirmation model of continuance intention to use mobile instant messaging. *Telematics and Informatics*, *33*(1), 34–47.

Oliver, R. (2014). *Satisfaction: A behavioral perspective on the consumer*. Routledge.

Ou, C., & Sia, C. (2010). Consumer trust and distrust: An issue of website design. *International Journal of Human Computing Studies*, *68*(12), 913–934.

Pal, D., Babakerkhell, M., & Zhang, X. (2021). Exploring the determinants of Users' continuance usage intention of smart voice assistants. *IEEE Access*, *3*, 162259–162275.

Parasuraman, A. (2000). Technology readiness index (TRI) a multiple-item scale to measure readiness to embrace new technologies. *Journal of Service Research*, *2*(4), 307–320.

Parasuraman, A., & Colby, C. (2015). An updated and streamlined technology readiness index. *Journal of Service Research*, *18*(1), 59–74.

Parasuraman, A., & Grewal, D. (2000). The impact of technology on the quality-value-loyalty chain: A research agenda. *Journal of the Academy of Marketing Science*, *28*(1), 168–174.

Park, D., & Namkung, K. (2021). Exploring users' mental models for anthropomorphized voice assistants through psychological approaches. *Applied Sciences*, *11*, 11147.

Park, H., & Zhang, Y. (2022). Technology readiness and technology paradox of unmanned convenience store users. *Journal of Retailing and Consumer Services*, *65*, 102523.

Pizzi, G., Vannucci, V., Mazzoli, V., & Donvito, R. (2023). I, chatbot! The impact of anthropomorphism and gaze direction on willingness to disclose personal information and behavioral intentions. *Psychology and Marketing*, *40*, 1372–1387.

Rauschnabel, P., & Ahuvia, A. (2014). You're so lovable: Anthropomorphism and brand love. *Journal of Brand Management*, *21*(5), 372–395.

Roy, S., Balaji, M., Quazi, A., & Quaddus, M. (2018). Predictors of customer acceptance of and resistance to smart technologies in the retail sector. *Journal of Retailing and Consumer Services*, *42*, 147–160.

Saeed, K. A., & Abdinnour-Helm, S. (2008). Examining the effects of information system characteristics and perceived usefulness on post adoption usage of information systems. *Information & Management*, *45*(6), 376–386.

Salles, A., Evers, K., & Farisco, M. (2020). Anthropomorphism in AI. *AJOB Neuroscience*, *11*(2), 88–95.

Sarstedt, M., Hair, J., Jr., Cheah, J., Becker, J., & Ringle, C. (2019). How to specify, estimate, and validate higher-order constructs in PLS-SEM. *Australasian Marketing Journal*, *27*, 197–211.

Schmidthuber, L., Maresch, D., & Ginner, M. (2020). Disruptive technologies and abundance in the service sector—Toward a refined technology acceptance model. *Technological Forecasting and Social Change*, *155*, 119328.

Schreibelmayr, S., & Mara, M. (2022). Robot voices in daily life: Vocal human-likeness and application context as determinants of user acceptance. *Frontiers in Psychology*, *13*, 787499.

Schweitzer, F., Belk, R., Jordan, W., & Ortner, M. (2019). Servant, friend or master? The relationships users build with voice-controlled smart devices. *Journal of Marketing Management*, *35*(7–8), 693–715.

Seeber, I., Bittner, E., Briggs, R., De Vreede, T., De Vreede, G., Elkins, A., Maier, R., Merz, A., Oeste-Reiß, S., Randrup, N., & Schwabe, G. (2020). Machines as teammates: A research agenda on AI in team collaboration. *Information & Management*, *57*(2), 103174.

Seo, S. (2022). When female (male) robot is talking to me: Effect of service robots' gender and anthropomorphism on customer satisfaction. *International Journal of Hospitality Management*, *102*, 103166.

Shank, D., Graves, C., Gott, A., Gamez, P., & Rodriguez, S. (2019). Feeling our way to machine minds: People's emotions when perceiving mind. *Computers in Human Behavior*, *98*, 256–266.

Söderlund, M., & Oikarinen, E.-L. (2021). Service encounters with virtual agents: An examination of perceived humanness as a source of customer satisfaction. *European Journal of Marketing*, *55*(13), 94–121.

Sohn, K., & Kwon, O. (2020). Technology acceptance theories and factors influencing artificial intelligence-based intelligent products. *Telematics and Information*, *47*, 101324.

Steinbruck, A. (2019). Personified machines—How voice assistants are anthropomorphised and what this has to do with our AI literacy: Medium. https://alexasteinbruck.medium.com/personified-machines-29875268f151

Stroessner, S., & Benitez, J. (2019). The social perception of humanoid and non-humanoid robots: Effects of gendered and machinelike features. *International Journal of Social Robotics*, *11*, 305–315.

Sweeney, J., & Swait, J. (2008). The effects of brand credibility on customer loyalty. *Journal of Retailing and Consumer Services*, *15*, 179–193.

Veeramootoo, N., Nunkoo, R., & Dwivedi, Y. (2018). What determines success of an e-government service? Validation of an integrative model of e-filing continuance usage. *Government Information Quarterly*, *35*(2), 161–174.

Voorhees, C., Brady, M., Calantone, R., & Ramirez, E. (2016). Discriminant validity testing in marketing: An analysis, causes for concern, and proposed remedies. *Journal of the Academy of Marketing Science*, *44*(1), 119–134.

Wixom, B., & Todd, P. (2005). A theoretical integration of user satisfaction and technology acceptance. *Information Systems Research*, *16*(1), 85–102.

Wolf, C., Joye, D., Smith, T., & Fu, Y. (2016). *The SAGE handbook of survey methodology*. SAGE.

Yang, G., Ji, G., & Tan, K. (2020). Impact of artificial intelligence adoption on online returns policies. *Annals of Operations Research*, *308*, 703–726.

Yilmaz, M., & Rızvanoğlu, K. (2021). Understanding users' behavioral intention to use voice assistants on smartphones through the integrated model of user satisfaction and technology acceptance: A survey approach. *Journal of Engineering, Design and Technology*, *20*(6), 1738–1764.

Yuan, S., Liu, L., Su, B., & Zhang, H. (2020). Determining the antecedents of mobile payment loyalty: Cognitive and affective perspectives. *Electronic Commerce Research and Applications*, *41*, 100971.

Zhu, Y., Wang, R., & Pu, C. (2022). "I am chatbot, your virtual mental health adviser." what drives citizens' satisfaction and continuance intention toward mental health chatbots during the COVID-19 pandemic? An empirical study in China. *Digital Health*, *8*, 1–15.

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14791838, 2024, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cb.2228, Wiley Online Library on [23/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

APPENDIX A

TABLE A1 Construct, definitions, items, and sources.

Construct and definition	Abbreviation	Items	References
Optimism Belief in the potential benefits of technology	OPT1	New technologies contribute to a better quality of life	Parasuraman and Grewal (2000); Parasuraman and Colby (2015)
	OPT2	Technology makes me more productive in my personal life	
	OPT3	Technology gives people more control in managing their daily lives	
	OPT4	Technology gives me better ability to move freely	
Innovativeness A tendency to be a technology pioneer and thought leader. Openness to adopting new and innovative technologies	INN1	I am among the first in my circle of friends/ families to acquire new technology when it appears	
	INN2	I can usually figure out new high-tech products and services without help from others	
	INN3	Other people come to me for advice on new technology	
	INN4	I keep up with the latest technological developments in my areas of interest	
Discomfort A perceived lack of control over technology and a feeling of being overwhelmed by it.	DIS1	Technical support lines are not helpful because they do not explain things in terms that I understand	
	DIS2	Sometimes, I think that technology systems are not designed for use by ordinary people	
	DIS3	There is no such thing as a manual for a high-tech product that is written in plain language	
	DIS4	When I get technical support from a provider of a high-tech product, I sometimes feel as if I am being taken advantage of by someone who knows more than I do	
Insecurity Distrust of technology, stemming from skepticism about its ability to work properly and concerns about potential harmful consequences.	INS1	People rely too much on technology to do things for them	
	INS2	Too much technology distracts people to a point that is harmful	
	INS3	Technology lowers the quality of relationships by reducing personal interaction	
	INS4	I do not feel confident doing business with a place that can only be reached online	
Confirmation Users' perception of the congruence between expectation of IS use and its actual performance.	CON1	My experience with it is better than expected	Bhattacherjee (2001)
	CON2	Its functionalities are better than expected	
	CON3	Overall, the majority of my expectations from using it are confirmed	
Satisfaction. Overall psychological state that emerges when emotions surrounding confirmed expectations are coupled with the users' prior feelings about the consumption experience.	SAS1	Overall, how satisfied are you with your digital assistant?	
	SAS2	Overall, how satisfied are you with your digital assistant?	
	SAS3	Overall, how satisfied are you with your digital assistant?	
	SAS4	Overall, how satisfied are you with your digital assistant?	
Use continuance intention. Users' intention to continued use of the IS.	INT1	I plan to continue using it	
	INT2	I will keep using it as regularly as I do now	
	INT3	My intention is to continue using it than using other alternatives	
	INT4	I intend to increase my use of it in the future	
Intelligence. The degree to which the users perceive the AI agent behavior to be efficient and autonomous with the ability to process and produce natural language and deliver effectual output.	PI1	It can complete tasks quickly	Moussawi et al. (2021)
	PI2	It can understand my commands	
	PI3	It can communicate with me in an understandable manner	
	PI4	It can find the necessary information for completing task	
	PI5	It can provide me with a useful answer	

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TABLE A1 (Continued)

Construct and definition	Abbreviation	Items	References
	PI6	It can process the necessary information for completing tasks	
Anthropomorphism. Degree to which a system is perceived as human-like.	PA1	It is able to speak like a human	
	PA2	It appears to be happy	
	PA3	It is friendly	
	PA4	It is respectful	
	PA5	It is caring	
Information accuracy. Factors which determine user's perception of quality of information, to which the users perceive the information is accurate, up to date and complete	INA1	The information provided is the correct information	Wixom and Todd (2005)
	INA2	The information provided has few errors	
	INA3	The information provided is accurate	
Information up-to-datedness: (recency)	INU1	The information provided is the most recent information	
	INU2	The information provided is always up to date	
	INU3	The information provided is the most currently accepted information	
Information completeness: (reliability)	INR1	It provides me with a complete set of information	
	INR2	It provides me with comprehensive information	
	INR3	It provides me with all the information I need	
System reliability: *Reliability* refers to the dependability of system operation.	SYR1	It operates reliably	
	SYR2	It performs tasks reliably	
	SYR3	It operates in a dependable manner	
System flexibility: *Flexibility* refers to the way the system adapts to changing demands of the users.	SYF1	It can adapt to meet a variety of needs	
	SYF2	It can flexibly adjust to new demands or conditions	
	SYF3	It can be versatile in addressing my needs as they arise	
System timeliness: *Timeliness* refers to the degree to which users perceive the system offers timely responses.	SYT1	It takes too long to respond to my requests (reverse coded)	
	SYT2	It provides information in a timely fashion	
	SYT3	It responses to my requests quickly	
Brand Trustworthiness: User's beliefs that the brand is honest, reliable, sincere, dependable, and guarantees the quality of their product.	BT1	Dependable (How do you perceive the brand?)	(Featherman et al., 2021)
	BT2	Honest (How do you perceive the brand?)	
	BT3	Reliable (How do you perceive the brand?)	
	BT4	Sincere (How do you perceive the brand?)	
	BT5	Trustworthy (How do you perceive the brand?)	
	BT6	Guarantees quality (How do you perceive the brand?)	
Brand Expertise: User's beliefs that the brand is knowledgeable and experienced, skilled, qualified, and innovative demonstrating genius engineering.	BE1	The brand is …an amateur…an expert	(Featherman et al., 2021)
	BE2	The brand is …not knowledgeable…knowledgeable	
	BE3	The brand is …not innovative…innovative	
	BE4	The brand …does not demonstrate genius engineering… demonstrates genius engineering	
	BE5	The brand is …not qualified…qualified	
	BE6	The brand is …not skilled… skilled	

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TABLE A2 Descriptive analysis of all survey items.

Construct	Abbreviation	Mean	SD
Innovativeness	INN1	4.46	1.619
	INN2	5.19	1.448
	INN3	4.60	1.592
	INN4	5.00	1.492
Optimism	OPT1	5.89	1.064
	OPT2	5.60	1.111
	OPT3	5.69	1.156
	OPT4	5.73	1.108
Insecurity	INS1	4.96	1.494
	INS2	4.93	1.594
	INS3	5.06	1.550
	INS4	3.86	1.678
Distrust	DIS1	3.76	1.472
	DIS2	3.51	1.650
	DIS3	3.81	1.630
	DIS4	3.65	1.744
Confirmation	CON1	5.19	1.120
	CON2	5.21	1.080
	CON3	5.30	1.097
Satisfaction	SAS1	5.49	0.919
	SAS2	5.40	0.995
	SAS3	5.42	0.998
	SAS4	5.47	0.993
Use continuance intention	INT1	5.56	1.197
	INT2	5.48	1.207
	INT3	5.25	1.228
	INT4	5.16	1.301
Information accuracy	INA1	5.43	1.067
	INA2	4.81	1.286
	INA3	5.23	1.092
Information up-to-datedness (recency)	INU1	5.64	1.077
	INU2	5.64	1.142
	INU3	5.53	1.072
Information completeness (reliability)	INR1	5.13	1.191
	INR2	5.23	1.125
	INR3	5.03	1.214
System reliability	SYR1	5.49	1.059
	SYR2	5.49	0.993
	SYR3	5.42	1.056
System flexibility	SYF1	5.12	1.179
	SYF2	5.07	1.259
	SYF3	5.06	1.222
System timeliness	SYT1	4.58	1.598
	SYT2	5.25	1.137
	SYT3	5.42	1.128

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TABLE A2 (Continued)

Construct	Abbreviation	Mean	SD
Anthropomorphism	PA1	4.94	1.469
	PA2	4.48	1.568
	PA3	4.95	1.432
	PA4	5.11	1.276
	PA5	4.50	1.386
Intelligence	PI1	5.53	1.076
	PI2	5.25	1.113
	PI3	5.38	1.119
	PI4	5.37	1.126
	PI5	5.34	1.135
	PI6	5.37	1.068
Brand trustworthiness	BT1	5.19	1.172
	BT2	5.10	1.177
	BT3	5.20	1.293
	BT4	4.96	1.217
	BT5	5.06	1.206
	BT6	5.21	1.407
Brand expertise	BE1	5.69	1.099
	BE2	5.80	1.053
	BE3	5.81	1.041
	BE4	5.75	1.051
	BE5	5.84	1.065
	BE6	5.83	1.009

Abbreviation: SD, standard deviation.

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